

Zhiyao He

zhiyaohe@outlook.com | <https://zhiyaohe.github.io>

Robot Learning · Robot Manipulation · Reinforcement Learning · Robotics Simulation · Computer Vision · LLM Agents

Profile

I am an AI engineer working on embodied AI systems. I build robot pipelines that connect perception, language models, planning, and control, from simulation (Gazebo, MuJoCo, Isaac Gym) to real hardware. I am seeking a PhD position in physical AI. I would love to spend the coming years exploring how robots can learn to perceive, reason, act, and endure reliably in real-world environments.

Education

M.S. in Signal and Information Processing Hohai University, Nanjing, China Sep 2015 – Jul 2018

Grade: 82.33 / 100 (research-based degree)

- Thesis work on machine learning for runoff forecasting; published two peer-reviewed journal papers and obtained one Chinese invention patent.
- Relevant courses: Information Acquisition and Processing, Optimization. Certificate: Deep Learning Specialization (DeepLearning.AI).

B.Eng. in Electrical Engineering and Automation Anhui Agricultural University, Hefei, China Sep 2011 – Jul 2014

CGPA: 3.65 / 4.00

- Finished the four-year program in three years, graduating one year early.
- Relevant courses: Control Theory, Circuit Design, C Programming, Digital Signal Processing.

A.S. in Computer Science University of the People, USA Nov 2021 – Nov 2024

CGPA: 4.00 / 4.00

- President's List for 12 terms.
- Relevant courses: Python, Java, Databases, Operating Systems, Software Engineering, Computer Networks.

Publications and Patent

- Zhiyao He, Mingxiang Yang, Hao Wang, et al. "[Annual Average Runoff Ensemble Forecast for Jinping I-Stage Hydropower Station Based on Elman Neural Network](#)," Water Resources and Power, 2017.
- Chenming Li, Lei Zhu, Zhiyao He, et al. "[Runoff Prediction Method Based on Adaptive Elman Neural Network](#)," Water, 2019.
- Zhiyao He (Patent holder). "Medium- and Long-term Runoff Forecasting Method Based on Mutual Information, Kernel PCA, and Elman Neural Network." [Chinese National Invention Patent, CN107463993B](#), granted 2020.

Work Experience

AI Engineer WellWin Technology Co., Ltd., Hengqin Island, China Mar 2026 – Present

- Built the perception, planning and control stack for voice- and text-controlled navigation and manipulation on the Unitree Go2-W quadruped robot with a D1 arm (details in Projects).
- Implemented the closed-loop control system of the AKI humanoid robot: a reinforcement learning policy layer, a joint target layer, and PD motor control.
- Designed the AKI robot's power management system, covering power input, distribution and conversion, and the load execution layer.
- Integrated YOLO-based perception with ASR, LLM and TTS modules for face recognition and voice interaction.
- Set up simulation environments and ran locomotion training for the AKI and Unitree G1/R1 humanoid robots in Gazebo, MuJoCo, and Isaac Gym.

Independent Study and Research Nanjing, China Jul 2019 – Feb 2026

- After Huawei, I planned to return to research. During this period my second journal paper (Water, 2019) was published and my invention patent was granted (2020).
- From 2021 to 2024, I completed a full computer science degree at University of the People (GPA 4.00/4.00, see Education) to retrain from signal processing toward AI and robotics. Part of this period overlapped with the COVID-19 lockdowns in China.

- Admitted to the M.Sc. in Data Science at Trinity College Dublin (2025 intake), but did not enroll.
- After that, I prepared for AI engineering interviews and joined WellWin Technology in March 2026.

Software Engineer Huawei Technologies Co., Ltd., Nanjing R&D Institute, China

Jul 2018 – Jun 2019

- Worked on computer vision systems for face and vehicle recognition, mainly on development and performance optimization.
- Debugged and tuned the systems together with other teams.

Research Intern China Institute of Water Resources and Hydropower Research, Beijing, China

Jun 2016 – Jun 2017

- Built a complete runoff prediction pipeline: data collection and cleaning, feature engineering, model development, hyperparameter tuning, training and evaluation.
- This work produced two journal papers and one invention patent (listed above).

Selected Projects

Voice-Controlled Navigation and Manipulation on Unitree Go2-W

- Perception: YOLO11 object detection with a wrist-mounted depth camera, plus 2D-to-3D coordinate transformation to locate the target in the world frame.
- Navigation: SLAM-based autonomous navigation to the target position.
- Language: a task planning pipeline with ASR, LLM, OpenClaw and TTS, so the robot can be controlled by natural language and reply by voice.
- Manipulation: a grasping pipeline based on forward/inverse kinematics and Jacobian-based motion control.
- Developed in ROS 2, tested in MuJoCo simulation and on the real robot.

AKI Humanoid Robot

- Real-time control loop for locomotion and interaction: an RL policy generates joint positions, a joint target layer does scaling, integration and clipping, and a PD layer drives the motors.
- ASR-LLM-TTS pipeline for natural voice interaction.

Voice RAG Assistant on AgiBot Lingxi X2

- An in-store retail assistant: customers ask about products, plans, and pricing by voice.
- Built a fully offline pipeline on a Jetson for events with unreliable networks: a bridge node streams the X2's microphone audio to a local FastAPI backend, which runs local ASR (FunASR/Paraformer-zh), a local LLM (Qwen2.5-7B via Ollama), pgvector RAG over the product database (PostgreSQL), and local TTS (Sherpa-ONNX / Piper).

Technical Skills

- Programming & Systems: Python, C++, Linux, Docker, Git
- Robotics: ROS/ROS 2, Isaac Gym, Gazebo, MuJoCo, SLAM, Navigation, Localization, Perception
- AI & Robot Learning: PyTorch, Machine Learning, Computer Vision, Reinforcement Learning, ASR, TTS, LLMs
- Robot Platforms: Unitree G1, Unitree Go2/Go2-W, AgiBot Lingxi X2

Research Interests

- Robot Learning and Reinforcement Learning
- Vision-Language-Action Models
- Robot Manipulation and Grasping
- Robotics Simulation, SLAM and Navigation

Honors and Awards

- President's List, University of the People (12 terms) | 2021–2024
- Outstanding Contribution Award, Huawei Nanjing R&D Institute | 2019
- Outstanding Graduate Student, Hohai University | 2017
- Hohai University Scholarship | 2015–2017
- Anhui Agricultural University Scholarship | 2012

Teaching and Outreach

Humanoid Robotics Curriculum on AgiBot Lingxi X2 WellWin Technology

- Designed a humanoid robotics curriculum for Macao secondary school students, progressing from no-code tools (LingXin, LinkCraft) to ROS 2 programming with the AimDK SDK, and finally to multi-platform integration.

Humanoid Robot Task Challenge, MYAIR 2027 Macau Youth AI and Robotics Competition

- Designed the task and rules for a humanoid robot ball-shooting competition, testing vision recognition, trajectory planning and 3D printing.

UoPeople Ambassador University of the People (AY2022, Term 5)

- Answered prospective students' questions about admissions and academics, helping applicants with limited educational opportunities gain access to quality higher education.